

Maxillary premolars

Maxillary first premolar		Maxillary second premolar
Mesial outline (buccal cusp)	Slightly concave Mesial slope: straight	The same
Mesial contact area	Just cervically to the occlusal/middle junction Centered buccolingually	The same
Distal outline (buccal cusp)	straight distal slope: convex	Straight Distal slope: straight
Distal contact area	The same as the mesial one or slightly more occlusal Buccally situated	The same or situated more cervically . buccally situated
<i>Outlines of lingual cusp</i>	All are convex N.B. the mesial and distal slopes are meeting in 90° angle	All are straight
buccal outline (proximal view)	Convex, with the maximum convexity... The buccal cusp tip is at the central line of the buccal root	The same
Lingual outline (proximal view)	Convex with the maximum convexity on the middle third. The cusp tip is in line with the lingual border of the lingual root	The same
The root	•buccal & lingual views: -buccal root: <i>pointed apex and may curve distally</i> -lingual root: <i>shorter with blunt apex</i> •proximal views: -developmental depression from the mesial contact area to a part of the root trunk -the buccal outline of the buccal root is straight with tendency to incline lingually While the lingual...buccally -the root trunk length is half the length of the whole root mesially. And takes only the apical third on the distal surface	•buccal & lingual views: Pointed apex •proximal views: The same as anterior teeth with shallow developmental grooves
Pulp cavity (chamber & canal)	•mesiodistal section: (similar to canines) - Has narrow pulp chamber pointed incisally -The root canal is narrow •labiolingual section: -shows wider pulp chamber (buccolingually) with two pulp horns -two root canals emerge from the floor of the chamber to open separately or at the same foramen •midroot cross section: -circular & centered in each root (or each side) -the lingual root canal is wider	The same Except that the root canals in labiolingual section emerge from the pulp chamber as one root canal then divides into two root canals which then reunite to open in the same foramen
features	-more prominent buccal ridge - buccal ridge is more prominent than the lingual one -more tapering from the contacts cervically on the buccal view (roughly trapezoid) -developmental grooves on both sides of the buccal ridge	-less prominent the same -less tapering The same

	-shorter lingual cusp -more observed lingual convergence -mesial cusp ridges are longer -mesial marginal ridge groove -mesial root depression and no distal one	-the two cusps are nearly equal -less observed -distal cusp ridges are longer -no mesial marginal groove -mesial and distal shallow developmental grooves
Occlusal aspect	-hexagonal -the two buccal sides are nearly equal -the mesial side is shorter than the distal side and may be concave -the mesiolingual is shorter than the distolingual -longer central groove and thus mesial & distal fossae are nearer to the margins -fewer supplemental grooves -the triangular ridge of the buccal cusp is more prominent than the lingual cusp's	-oval (more rounded angle) -shorter central groove and thus mesial & distal fossae ... -more supplemental grooves - the same -the crown and root are thick at the cervix

By Mahmoud Shebl